

3.6 (Cont'd)

With the load coil connected to the R. F. output terminals put the load (piece to be heated) in position. Set the tank coil tap to turn No. 6 as a preliminary adjustment. The tank coil is on the upper shelf and has a moveable link for tap connections numbered one through sixteen from left to right. Set the "GRID CONTROL" knob on the front of the generator to about 30 and turn the "POWER CONTROL" knob to zero.

The generator is equipped with a 45 second time delay relay to allow for proper warm-up of the oscillator tube filament after the "FILAMENT-ON" button has been pushed. After this period, the red "OVERLOAD/RESET" lamp will light. Now press "OVERLOAD/RESET" button which readies the plate control circuit and the red pilot lamp will go out. Next press the "PLATE-ON" pushbutton to energize the plate contactor. (Generators with primary saturable reactor or control diode power control do not have "PLATE-ON" and "PLATE-OFF" pushbuttons). To apply RF power press the "RF POWER-ON" pushbutton. The initial meter readings should be very low. Advance the "POWER CONTROL" knob to 50. At this point, both grid milliamperes and plate meters should be reading and both should increase in proportion to increases of the "POWER CONTROL" setting.

IMPORTANT: If the three phases of the AC power supply have not been connected in the proper sequence the "PLATE-KV" meter will not go smoothly from zero to maximum as the power control knob setting is increased. If this happens, interchange any two of the three supply line conductors at the external circuit breaker. (This does not apply if the generator is equipped with primary saturable reactor or control diode power control).

3.6 (cont'd)

If grid current stays at zero or is very low and decreasing turn the "GRID CONTROL" knob to a higher number.

The variables tabulated below will affect grid and plate current as shown:

Variable	Effect On:	
	Grid Milliamperes	Plate Amperes
(a) Turn "GRID CONTROL" knob to higher number	increase	increase (slightly)
(b) Move tank tap to a higher number	decrease	increase
(c) Increase number of turns of work coil	decrease	increase
(d) Move work coil closer to workpiece	decrease	increase
(e) Increase workpiece diameter or resistivity	decrease	increase

Maximum loading has been obtained when the "PLATE AMPERE" meter reading is at the red line when the "POWER CONTROL" is set at 100. This condition will give maximum power output. Observe the "PLATE AMPERE" meter reading, and if more or less plate current is desired, move the tap to higher or lower numbered turns, respectively. It is, of course, necessary to turn off plate power and keep it off while making tap changes.

If, at any time, the plate current should exceed the limit, due to over-coupling, the plate overload relay will function and the red, "OVERLOAD/RESET" pilot light will come on. Reset this

3.6 (cont'd)

relay by pushing the "OVERLOAD/RESET" button. Decouple the tank coil by moving the tank tap back to a lower turn and try again. During this procedure, it is important to watch the grid current meter at the same time. For proper performance, the grid current should be adjusted to approximately 225 MA when set for maximum power and maximum plate current.

NOTE: Plate current should track with grid current. If not, then percentage of grid current must be increased. (See also "Grid Current" paragraph 3.7 following.) The grid is protected by an overload relay which will trip if grid current is excessive. If this trips, reset the circuit by pushing the "OVERLOAD/RESET" button.

It should be noted that in some applications optimum loading and maximum generator output may be neither practical nor desirable. Three possible conditions are:

(1) If optimum loading has been achieved by selection of tank taps, heating rate may be decreased either by turning down the "POWER CONTROL" knob or by moving the tank tap to lower numbers (from right to left). The latter approach is preferable to using a very low setting of the power control knob.

(2) If coil design and workpiece are such that it is not possible to reach red line plate current, but heating rate is satisfactory at a "POWER CONTROL" setting of 100 or less, operation should be satisfactory.

3.6 (cont'd)

(3) If the plate current reaches the red line value at a "POWER CONTROL" setting of less than 100, the tank coil tap setting is incorrect and must be moved one or more turns down to a lower number (from right to left).

3.7 GRID CURRENT

The recommended range of grid current for operation at reduced plate voltages are presented in Figure 1. Grid current can be adjusted to the desired value by means of the "GRID CONTROL" knob on the front panel. The adjustment may be either with plate power on or off. If the current should exceed the meter red line value the grid current overload relay, K₁, will trip and switch off plate power.

If the grid current is too low the oscillator tube will rectify which is detrimental and must be watched for when setting up a new job. In this case, the grid current will be zero or drifting down towards zero while plate current keeps rising. If this happens, shut off the plate power immediately and increase the "Grid Control" setting for higher grid current.

In many applications grid and plate current will vary with the temperature of the material to be heated. This is particularly noticeable when heating magnetic materials, e.g. steel, through the Curie temperature. In this case, the plate current shows a high reading when starting, which drops to a lower value as the load reached the Curie temperature. The grid current reacts inversely, starting with a low

3.7 (Cont'd)

reading and increasing to a maximum when above the Curie temperature. In this case select a grid setting which will keep the excursion of the grid current within the recommended range in Figure 1.

If the load is very light, for example with plate current one half or less of the value permitted in Figure 1, a grid current near the minimum permissible value is suggested for best tube life.

3.8 LOAD COILS

The oscillator circuit of this generator is so designed that matching transformers within the generator are not normally required. This permits the use of long load coil leads where required by the application. However, as a general rule, these leads should be kept as short as possible to prevent unnecessary losses and radiation. Lepel coaxial leads are recommended where long leads are required.

Load coils may be constructed of 1/8" to 3/8" copper tubing, 1/4" tubing being the size most commonly used. In general, load coils should be so designed that the coil surrounds the part of section to be heated with as many turns as possible, while the clearance between i.e. of coil and o.d. of load is kept to a practical minimum. Only in this manner is maximum efficiency and energy transfer to the load possible.

3.8 (cont'd)

On applications where the area of the part to be heated is restricted and only a few load coil turns are possible, an RF current transformer may be used. The standard Lepel load coil transformer is designed to accommodate load coils from the very smallest up to 2 uH with excellent efficiency. For special load coil design on unusual applications, the Engineering Department of Lepel High Frequency Laboratories, Incorporated may be consulted.

3.9 POWER CONTROL

Lepel generators can be ordered with a variety of optional power controls. The type provided in a specific generator can be determined by referring to the schematic diagram or by the code letter after the generator serial number on the nameplate. The most common types with the corresponding code letters are:

Primary Saturable Reactor with Tube Type Rectifier	(E)
Primary Saturable Reactor with Silicon Diode Rectifier Stacks	(J)
Thyratron Regulator	(RP)
Primary SCR Control with Silicon Diode Rectifier Stacks	(T)
Power Control Diode with Silicon Diode Rectifier Stacks	(S)
Power Control Diode with Tube Type Rectifier	(AA)

3.9.1 PRIMARY SATURABLE REACTOR WITH TUBE TYPE RECTIFIER (E)

This type of control provides smooth continuous control of power with very low ripple modulation of the RF output. The control range does not extend to zero and response time is generally not fast enough for very short heat cycles e.g. several seconds or less.

The reactors function as variable impedance in series with the three phase AC input to the plate transformer. When fully saturated by a DC current applied to the reactor control windings, impedance is minimum and generator output is maximum.

3.9.1 (Cont'd)

The DC current for the reactor control windings may be supplied by either a manually controlled (variable autotransformer) DC supply or by an SCR driver with provisions for automatic control. The applicable type will be indicated on the generator schematic diagram. In the automatic mode the SCR driver is controlled by a low level DC signal, typically zero to five milliamperes. This signal may be provided by an external source such as a temperature controller. See the paragraph dealing with temperature control in section 3.5.

The high voltage AC from the plate transformer is rectified by six mercury vapor tubes in a three-phase full wave bridge circuit.

3.9.2 PRIMARY SATURABLE REACTOR WITH SILICON DIODE RECTIFIER STACKS (J)

This system is identical to that of the preceeding section (section 3.7.1) except that the high voltage rectifier utilizes six silicon diode stacks. The combination of reactors and solid state devices is exceptionally reliable.

3.9.3 THYRATRON REGULATOR (RP)

This Lepel T-660 Thyatron Regulator provides smooth continuous control from almost zero power to 100% power. Response time is very short permitting heat cycles as short as a few hundreths of a second. The generator control circuit is arranged so that the heavy duty plate contactor may be energized continuously and the RF power rapidly switched on and off by the T-660 unit.

3.9.3 (Cont'd)

Both manual and automatic modes are possible as described in section 3.5. With manual control power output is proportional to the setting of the "POWER CONTROL" potentiometer. With automatic control power output will be proportional to an external DC current signal applied to the input of the T-660.

The T-660 regulator adjusts the timing (or phase shift) of the triggering pulses applied to the grids of the thyatron tubes, V1, V2 and V3. This increases or decreases the DC plate voltage applied to oscillator tube and hence the power output of the generator.

Control will respond smoothly from zero to maximum only if the three-phase AC power is connected in the proper phase sequence. If at initial start-up it does not respond smoothly interchange any two of the three feed conductors at the external disconnect switch.

See Lepel manual #13190002 for details on operation and repair of the T-660 regulator.

3.9.4 PRIMARY SCR CONTROL WITH SILICON DIODE RECTIFIER STACKS (T)

The Primary SCR control is a completely solid state system which provides continuous control from zero to 100% percent, very fast response and electronic over current protection. The generator control circuits are arranged so that the heavy duty plate contactor may be energized continuously while RF power is rapidly switched on and off

3.9.4. (Cont'd)

electronically by the primary SCR unit. Heating cycles of several hundredths of a second are possible.

Integrated circuits in the control unit adjust the timing (or phase shift) of gate pulses applied to the silicon controlled rectifiers (SCR's) in the power control. This varies the AC voltage applied to the primary of the plate transformer and consequently the RF power output.

Both manual and automatic modes are possible as described in Section 3.5. With manual control power output is proportional to the setting of the "POWER CONTROL" potentiometer. With automatic control power output will be proportional to an external DC current signal applied to the input of the controller.

If the AC power conductors are not connected in the correct phase sequence the controller will cut power to zero regardless of the control potentiometer setting. The "RF POWER" lamp will turn on, even though power is off, with improper phasing. This is corrected by interchanging any two of the AC feed cables at the external disconnect switch.

High voltage silicon diode rectifier stacks in a three phase, full wave, bridge connection provide rectification for the DC plate power supply.

3.9.5 POWER CONTROL DIODE WITH SILICON DIODE RECTIFIER STACKS (S)

The power control diode system provides smooth continuous control from minimum to maximum power with very low ripple modulation